Semester: 6th
Subject Name:- Machine Learning & Code:- CS31002
Branch (s): - CSE

SPRING MID SEMESTER EXAMINATION-2026

School of Computer Engineering
Kalinga Institute of Industrial Technology, Deemed to be University
Subject: Machine Learning
[Code: CS31002]

Time: 1 1/2 Hours **Full Mark: 20**

Answer Any four questions including question No.1 which is compulsory.
The figures in the margin indicate full marks.
Candidates are required to give their answers in their own words as far as practicable and all parts of a question should be answered at one place only.

1. Answer all the questions. [1 Mark × 5]

a) What does the complementary slackness condition in KKT imply for support vectors in a hard-margin SVM?

b) Given the data values: X={20, 40, 60, 80}. Normalize the data using Min–Max Normalization to the range [-1,1].

c) Imagine you are building a binary classification model to predict whether a customer will buy a product (1 = "yes", 0 = "no"). For a particular customer, the model predicts a probability of 0.7 for buying the product. However, the customer does not buy the product (true label = 0). Using binary log loss, calculate the loss for this prediction.

d) In KNN classification how to select a suitable value of K, and what would be the effect of choosing a very small value of K?

e) Draw a sketch illustrating the bias–variance trade-off in a machine learning model.

2. Derive the mathematical expression for computing the model parameters (weights) in Logistic Regression using the cross-entropy loss function. Additionally, write the Pseudocode for training a Logistic Regression model. [5 Marks]

3. (a) Construct the decision tree using ID3 for the above data set. Then predict the class label for: [Store Location = Suburban, Festival Offer = Yes, Product Category = Grocery]. [3 Marks]

Store Location	Festival Offer	Product Category	Sales Trend (Class)
Urban	Yes	Electronics	Down
Urban	No	Electronics	Down
Urban	No	Grocery	Down
Suburban	Yes	Electronics	Down
Suburban	Yes	Grocery	Down
Suburban	No	Grocery	Up
Suburban	No	Electronics	Up
Rural	Yes	Electronics	Up
Rural	No	Grocery	Up
Rural	No	Electronics	Up

(b) Critically analyze the differences among Batch Gradient Descent, Stochastic Gradient Descent, and Mini-Batch Gradient Descent. [2 Marks]

4. (a) A multi-class classification model is evaluated on a dataset containing three classes: C1, C2, and C3. The resulting confusion matrix of the model is shown below, where rows represent actual labels and columns represent predicted labels. Compute the accuracy, and micro-average, macro-average and weighted average of precision, recall and f1-score. [3 Marks]

		Predicted		
		C1	C2	C3
Actual	C1	2	1	0
	C2	0	1	0
	C3	1	2	3

(b) Write the mathematical expression for the Ridge Regression estimator of the regression coefficients. [2 Marks]

5. (a) In a sample observation, there are two independent predictors X_1 and X_2, and a dependent target variable Y. The observed data are given in the table below:

X_1	X_2	Y
1	2	6
2	1	5
3	4	10
4	3	9
5	5	13

Fit a straight line $\mathbf{y = \beta_0 + \beta_1 X_1 + \beta_2 X_2}$ to the data by the method of least squares. Predict the value of the target variable Y for a new sample where $X_1 = 6$ and $X_2 = 3$. [3 Marks]

(b) Apply Naive Bayes Classifier to predict the class for the instance [Red, Circle, Medium][2 Marks]

Color	Shape	Size	Class
Red	Circle	Small	0
Blue	Square	Medium	1
Green	Circle	Large	0
Red	Triangle	Medium	1
Blue	Circle	Small	0
Green	Square	Medium	1

---: Best of Luck:---